

Developing simulation and statistical evaluation tools for a nonlinear mixed-effects model

Extending and validating the joint model in Leaspy

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CentraleSupélec

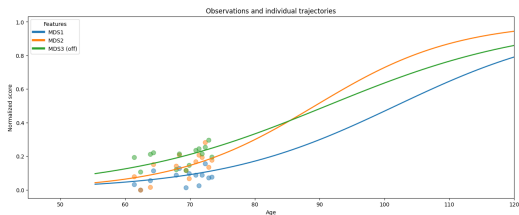


Section 1

Motivation and context

Longitudinal data: repeated observations over time

- The **same patient i** is observed at several chronological ages t .
- Visits can be sparse, irregular, unbalanced, and partially missing.
- Repeated measurements from one patient are correlated.



Dots: observations at successive visits. Curves: personalised trajectories.

Nonlinear mixed-effects models combine a population trajectory with patient-specific deviations.

Reconstruct disease dynamics from short, heterogeneous follow-ups.

Why simulation is indispensable here

Known truth

Real cohorts never reveal
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Support reproducibility and benchmarking without distributing patient records.

Internship gap. The joint-model paper used a separate research script; Leaspy's standard `simulate` interface supported only the logistic model.

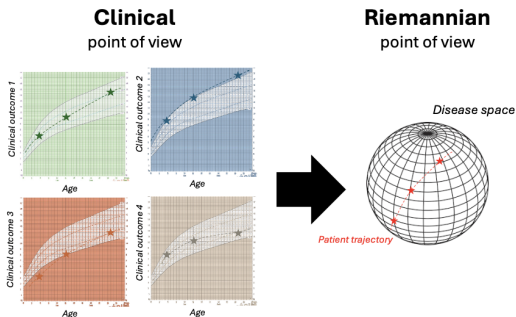
Goal: integrate joint simulation in Leaspy, then test the complete simulate–fit–personalise loop.

Simulation-study principles: Morris et al. [1].

Section 2

How Leaspy represents disease progression

From separate scores to a disease-space trajectory



- **Clinical view:** one curve per outcome against age.
- **Riemannian view:** their joint evolution is one path γ_i in disease space.

Leaspy represents several clinical scores through bounded logistic trajectories.

Figure: Ortholand [2]; disease course mapping: Schiratti et al. [3].

Nonlinear mixed effects: population and individual

Fixed effects:
shared by the cohort
 $(g, v_0, t_0, \sigma_\tau, \sigma_\xi)$

Random effects: one set per patient
 $(\tau_i, \xi_i, \mathbf{w}_i, \zeta_i)$

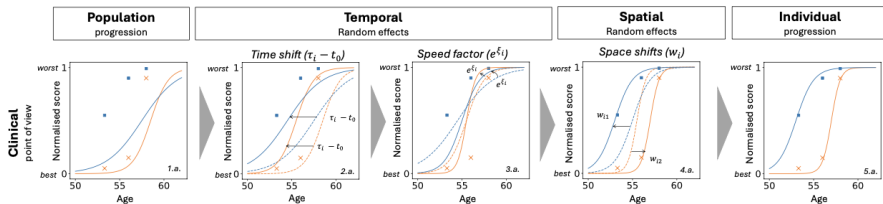
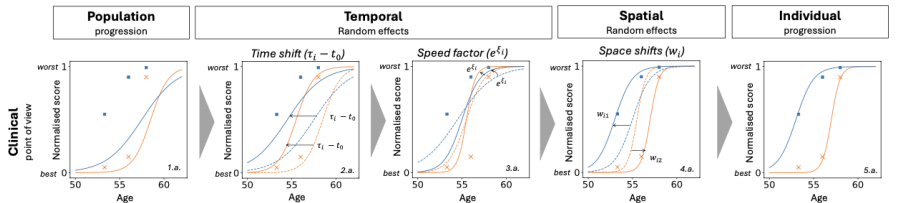


Figure adapted from Koval et al. [4].

Disease age separates disease stage from chronological age

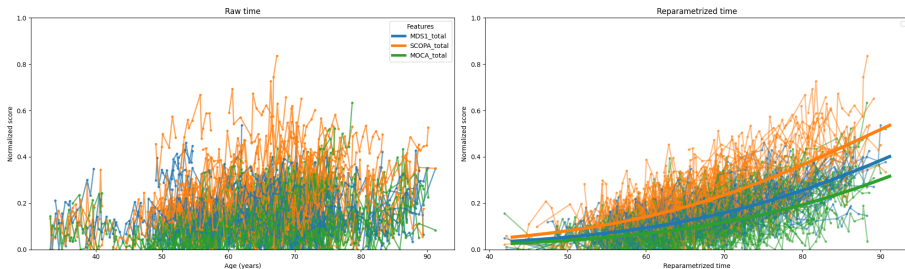


$$\psi_i(t) = e^{\xi_i}(t - \tau_i) + t_0$$

$\psi_i(t)$ is the latent disease age of patient i at chronological age t ; τ_i is the individual reference age and ξ_i the log-acceleration.

Figure adapted from Koval et al. [4].

How random effects personalise the trajectory



$$\tau_i$$

individual reference age;
shifts the trajectory
earlier or later

$$e^{\xi_i}$$

accelerates or slows
disease progression

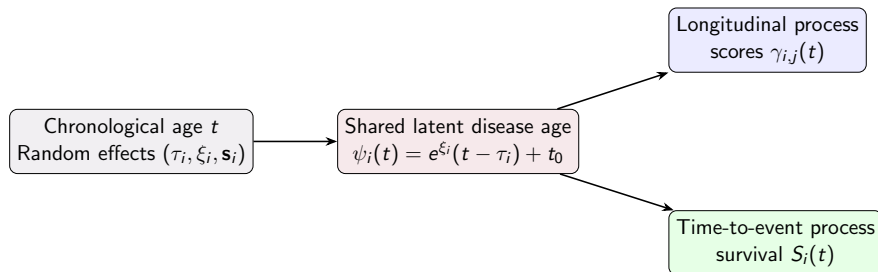
$$t_0$$

anchors the population
disease timeline

Time shift, acceleration, and space shifts transform one population curve into an individual course.

Individual disease-course transformations: Koval et al. [4].

The joint model: one disease age, two processes



- **Competing events:** several mutually exclusive outcomes can end follow-up.
- **Informative and non-informative dropout:** follow-up can end for a reason related to the patient's unobserved health state or for a reason unrelated to the disease.

Joint latent-disease-age models: Ortholand et al. [5, 6].

Section 3

How the cohorts are simulated with the joint model

Two input layers must remain distinct

Learned by fit

- longitudinal shape: $t_0, \mathbf{g}, \mathbf{v}_0$
- heterogeneity: σ_τ, σ_ξ
- observation noise
- survival: ν, ρ
- source mixing: $\mathbf{A}, \zeta$

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Specified by the user

- cohort size N
- first-visit offset: $\bar{\delta}_f, \sigma_{\delta_f}$
- follow-up: $\bar{T}_f, \sigma_{T_f}$
- inter-visit gap: $\bar{\delta}_v, \sigma_{\delta_v}$
- minimum visit spacing
- or an explicit DataFrame visit schedule

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The model determines what disease progression looks like; the user determines what the study looks like.

Simulation algorithm (1/2): create each follow-up

For each patient $i = 1, \dots, N$:

1. Random effects

draw patient-specific effects

$$\xi_i \sim \mathcal{N}(0, \sigma_\xi^2)$$

$$\tau_i \sim \mathcal{N}(t_0, \sigma_\tau^2)$$

plus latent sources, when
present

2. First visit

draw the first-visit time

$$t_{i,0} = \tau_i + \delta_{f_i}$$

$$\delta_{f_i} \sim \mathcal{N}(\bar{\delta}_f, \sigma_{\delta_f}^2)$$

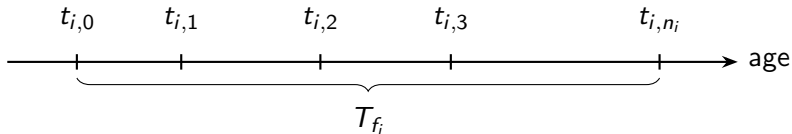
3. Visit schedule

draw inter-visit gaps

$$\delta_{v_{i,r}} \sim \mathcal{N}(\bar{\delta}_v, \sigma_{\delta_v}^2)$$

until the follow-up duration

$$T_{f_i} \sim \mathcal{N}(\bar{T}_f, \sigma_{T_f}^2)$$



Steps 2–3 are governed by the user-defined study design.

Simulation algorithm (2/2): outcomes and events

For each patient $i = 1, \dots, N$:

4. Outcomes

compute disease age and trajectory

$$\psi_i(t_{i,r}), \gamma_i$$

draw bounded observations with Beta noise

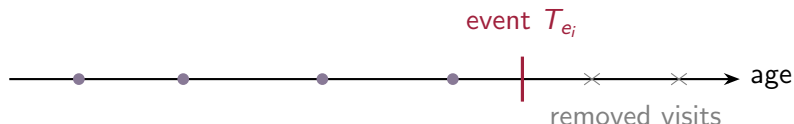
5. Event time

inverse-transform sampling from survival

$$T_{ei} \sim e^{-\xi_i} \mathcal{W}(\nu, \rho) + \tau_i$$

6. Censoring

remove visits after T_{ei}
right-censor events beyond the final visit



Output: longitudinal observations, an event or censoring time, and its indicator.

Integrating joint simulation inside Leaspy

1. Standard API

Extend estimate with longitudinal and event predictions, so joint models use the usual simulate workflow.

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Accept a random visit process or an explicit DataFrame; infer missing random-mode study parameters from an input dataset.

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Draw Beta observations; clamp infeasible variance to $0.99 \mu(1 - \mu)$ and warn about the affected observation.

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4. Single and competing events

Evaluate conditional survival/CIFs on the visit grid; sample the event by inversion and its cause from incremental cause-specific CIFs.

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5. Rounding-safe censoring

Round visit times, right-censor events beyond follow-up, then remove any rounded visit that falls after the event time.

One Leaspy workflow now produces bounded trajectories, event types, and coherent censoring.

Section 4

How the self-consistency of the simulation is tested

How simulation quality is assessed

Reference model
population truth θ^*

What is simulated exactly?

Reference: multivariate joint model fitted on the PRO-ACT ALS dataset
 $J = 4$ longitudinal features, $K = 2$ latent sources, $L = 1$ event (death)

Bulbar function

Speech, salivation, and swallowing items.

Gross-motor function

Turning in bed; walking; climbing stairs.

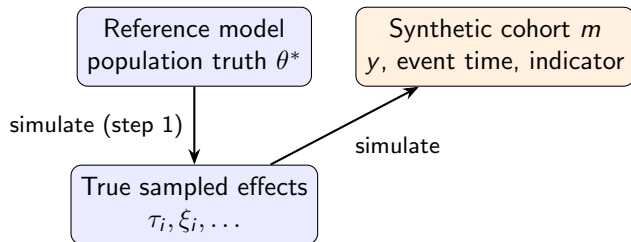
Fine-motor function

Handwriting; cutting food/handling utensils; dressing and hygiene.

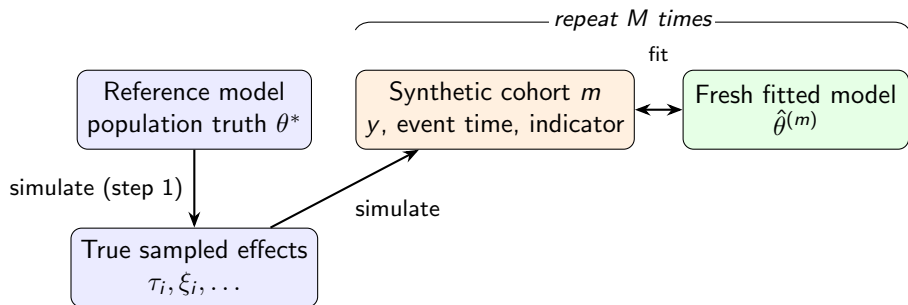
Total function

Overall ALSFRS-R: all items, including the preceding domains and respiratory items.

Step 1: simulate from a known reference model



Step 2: fit a fresh model to each synthetic cohort



Experimental setting: deliberately information-rich

Replications

$N = 1000$ patients

$M = 200$ cohorts

Visit process

first offset $-2 \pm 1y$

follow-up $6 \pm 2y$

interval $0.083 \pm 0.042y$

minimum 0.05y

Estimation

MCMC-SAEM

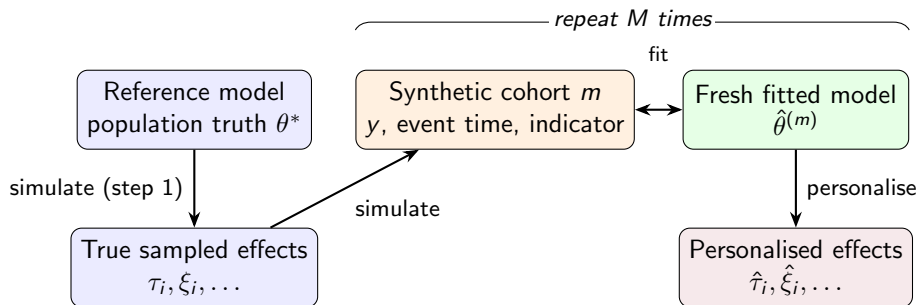
70 000 fit iterations

5 000 personalisation
iterations

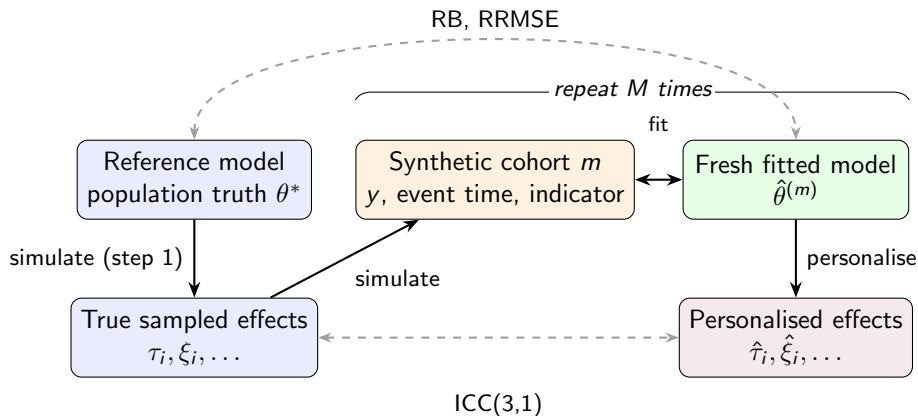
Interpretation boundary: short visit intervals and long follow-up create a favourable dense-observation setting.

MCMC-SAEM for nonlinear mixed-effects models: Kuhn et al. [7].

Step 3: retain and recover individual effects



Step 4: compare population and individual recovery



Three recovery metrics, two inference levels

Fixed effects / population parameters

Define the relative error in repetition m as

$$e_m = \frac{\hat{\theta}^{(m)} - \theta^*}{|\theta^*|} \times 100.$$

$$\text{RB} = \frac{1}{M} \sum_{m=1}^M e_m$$

signed systematic error.

Best value: 0%.

$$\text{RRMSE} = \sqrt{\frac{1}{M} \sum_{m=1}^M e_m^2}$$

total error from bias and dispersion.

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Individual random effects

Role in the pipeline. Compute one ICC per random effect and cohort, comparing true sampled with personalised values.

ICC(3,1): two-way mixed, consistency, single measure

Patients are the targets. True and personalised values are the two fixed measurement methods ($k = 2$).

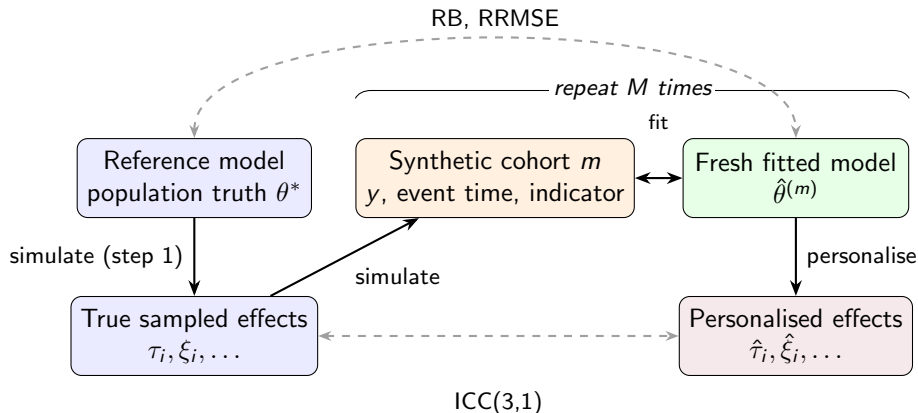
$$\text{ICC}(3, 1) = \frac{MS_B - MS_E}{MS_B + (k - 1)MS_E}$$

MS_B : between-patient variation.

MS_E : residual true–estimate disagreement.

Best value: 1 (perfect agreement).

The complete self-consistency experiment



Self-consistency asks whether the model can recover what it generated.

Section 5

Results of the experiments

Fixed effects / population parameters: recovery (1/2)

Parameter	Interpretation	RB (%) [95% CI]	RRMSE (%) [95% CI]
t_0	Population reference time	0.44 [0.35, 0.53]	0.83 [0.76, 0.90]
σ_τ	Time-shift SD	0.08 [-0.24, 0.38]	2.32 [2.10, 2.54]
σ_ξ	Log-acceleration SD	0.03 [-0.27, 0.37]	2.32 [2.11, 2.56]
ρ	Weibull shape	0.10 [-0.61, 0.81]	5.01 [4.50, 5.51]
ν	Weibull scale	2.33 [1.64, 3.00]	5.43 [4.97, 5.91]
$\sigma^{(0)}$	Bulbar observation-noise SD	-4.60 [-4.76, -4.45]	4.73 [4.58, 4.89]
$\sigma^{(1)}$	Fine-motor observation-noise SD	-4.16 [-4.28, -4.04]	4.25 [4.14, 4.38]
$\sigma^{(2)}$	Gross-motor observation-noise SD	-3.08 [-3.17, -2.97]	3.17 [3.07, 3.27]
$\sigma^{(3)}$	Total-score observation-noise SD	-1.26 [-1.37, -1.15]	1.47 [1.37, 1.56]

Best recovered: temporal population parameters, supported by many repeated observations per patient.

Fixed effects / population parameters: recovery (2/2)

Parameter	Interpretation	RB (%) [95% CI]	RRMSE (%) [95% CI]
$g^{(0)}$	Bulbar trajectory position	-8.27 [-9.41, -7.23]	11.41 [10.60, 12.28]
$g^{(1)}$	Fine-motor trajectory position	-8.80 [-10.03, -7.52]	12.61 [11.55, 13.66]
$g^{(2)}$	Gross-motor trajectory position	-7.60 [-8.71, -6.52]	11.16 [10.26, 12.06]
$g^{(3)}$	Total-score trajectory position	-5.92 [-6.68, -5.17]	8.09 [7.46, 8.76]
$v_0^{(0)}$	Bulbar trajectory velocity	6.46 [5.32, 7.65]	10.53 [9.56, 11.53]
$v_0^{(1)}$	Fine-motor trajectory velocity	2.23 [1.54, 2.93]	5.57 [5.05, 6.12]
$v_0^{(2)}$	Gross-motor trajectory velocity	1.12 [0.52, 1.74]	4.52 [4.12, 4.92]
$v_0^{(3)}$	Total-score trajectory velocity	2.05 [1.37, 2.73]	5.25 [4.75, 5.78]

Hardest to recover: trajectory positions g and bulbar velocity $v_0^{(0)}$, which can compensate for each other.

Individual random effects: recovery (1/2)

Effect	ICC(3,1)
ξ_i	0.988 ± 0.004
τ_i	0.995 ± 0.002
$w_{i,0}$	0.987 ± 0.006
$w_{i,1}$	0.989 ± 0.003
$w_{i,2}$	0.990 ± 0.003
$w_{i,3}$	0.987 ± 0.003
$(\zeta^\top \mathbf{s}_i)_0$	0.965 ± 0.027

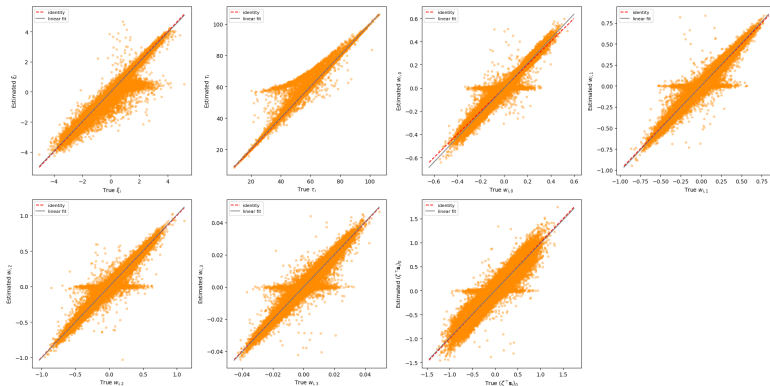
Mean $\pm$ standard deviation across repetitions.

Strong agreement for every evaluated individual effect: all ICCs exceed 0.96.

The **survival shift** is least precise: it is informed by at most one possibly censored event per patient.

Source coordinates $\mathbf{s}_i$ are rotation-dependent, so invariant feature shifts $\mathbf{w}_i = \mathbf{A}\mathbf{s}_i$ and survival shifts are compared instead.

Individual random effects: recovery (2/2)



Aggregate ICC is high, but some low-information patients are pulled toward the population mean.

True sampled values (x-axis) vs personalised estimates (y-axis). Dashed red: identity; grey: linear fit. Points are pooled over patients and repetitions.

Section 6

Conclusions

Take-home messages

Goal: integrate joint simulation in Leaspy, then test the complete simulate–fit–personalise loop.

- 1 **Integration achieved:** joint simulation now uses the standard Leaspy API, including survival, censoring, and competing events.
- 2 **Validation achieved:** the complete loop is self-consistent in the favourable dense setting studied.
- 3 **Main result:** temporal and individual effects are recovered very well; trajectory positions and survival-related quantities are less precisely identified.
- 4 **Scope:** recovery under sparse visits, heavier censoring, or model misspecification remains to be evaluated.
- 5 **Future work:** the same evaluation pipeline will certainly be used during my PhD on "*Explainability and variable selection for multivariate longitudinal data*" as many methods will be developed and tested on synthetic data before being applied to real clinical datasets.

Patient privacy

Synthetic cohorts do not correspond to real patients and can facilitate sharing.

They still require clear labelling, provenance, governance, and disclosure-risk assessment.

Computational cost

200 fits $\times$ 70 000 iterations plus personalisation is not negligible.

Small preliminary runs and saved intermediates avoid unnecessary recomputation.

Software responsibility

Errors can propagate into scientific conclusions.

Unit tests, functional tests, documentation, and peer review are part of methodological rigour.

Synthetic does not mean risk-free, and reproducible does not mean cost-free.

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Appendix: notation and normalised disease space

$i = 1, \dots, N$ patient

$r = 1, \dots, n_i$ visit

$j = 1, \dots, J$ longitudinal feature

$k = 1, \dots, K$ latent source

$l = 1, \dots, L$ event type

- $t_{i,r,j}$: chronological age
- $y_{i,r,j}$: observed score
- $\gamma_{i,j}$: latent individual trajectory
- $\epsilon_{i,r,j}$: observation noise

With J normalised clinical scores, the disease space is $(0, 1)^J$. The chosen metric produces bounded logistic trajectories in this space.

$$y_{i,r,j} = \gamma_{i,j}(t_{i,r,j}) + \epsilon_{i,r,j}$$

Appendix: why the metric produces a logistic trajectory

1. Flatten the Riemannian geometry

For one score, let $M = (0, 1)$ with

$$ds^2 = G(p) dp^2, \quad G(p) = \frac{1}{p^2(1-p)^2}.$$

Define the logit coordinate

$$\phi(p) = \log \frac{p}{1-p}, \quad \phi'(p) = \frac{1}{p(1-p)}.$$

Since $\phi'(p)^2 = G(p)$, we obtain $ds^2 = d\phi^2$: the geometry is Euclidean in the ϕ coordinate.

2. Geodesics become straight lines

Therefore a geodesic γ satisfies

$$\phi(\gamma(t)) = \phi(p_0) + c(t - t_0).$$

$$\gamma(t) = \left[1 + g \exp\left(-\frac{(1+g)^2}{g} v_0(t - t_0)\right) \right]^{-1}$$

3. Map back to $(0, 1)$

Applying the inverse logit gives

$$\gamma(t) = \left[1 + \frac{1-p_0}{p_0} e^{-c(t-t_0)} \right]^{-1},$$

which is a bounded logistic trajectory.

4. Recover Leaspy's parametrisation

Set $p_0 = 1/(1+g)$ and require $\dot{\gamma}(t_0) = v_0$.

Because $\dot{\gamma} = c\gamma(1-\gamma)$,

$$c = \frac{v_0}{p_0(1-p_0)} = \frac{(1+g)^2}{g} v_0.$$

A straight geodesic in logit coordinates becomes a logistic curve in score space.

Appendix: full longitudinal model

$$\left\{ \begin{array}{l} \psi_i(t) = e^{\xi_i}(t - \tau_i) + t_0, \\ \mathbf{w}_i = \mathbf{A}\mathbf{s}_i, \\ \gamma_{i,j}(t) = \left[1 + g_j \exp\left(-\frac{(1 + g_j)^2}{g_j} (v_{0,j}(\psi_i(t) - t_0) + w_{i,j})\right) \right]^{-1}, \\ y_{i,r,j} = \gamma_{i,j}(t_{i,r,j}) + \epsilon_{i,r,j}, \quad \epsilon_{i,r,j} \sim \mathcal{N}(0, \sigma_j^2). \end{array} \right.$$

- $p_j = 1/(1 + g_j)$ is the population trajectory value at t_0 .
- $v_{0,j}$ controls the velocity at the reference time.
- $w_{i,j}$ changes the relative ordering of feature trajectories.

Appendix: survival and latent-source coupling

Single-event survival on disease age

$$S_i(t) = \mathbf{1}_{\psi_i(t) > t_0} \exp \left[- \left(\frac{\psi_i(t) - t_0}{\nu} \right)^\rho \right] + \mathbf{1}_{\psi_i(t) \leq t_0}.$$

Event-specific scale with sources

For event l ,

$$\nu_{i,l} = \nu_l \exp \left[-\xi_i - \frac{1}{\rho_l} (\boldsymbol{\zeta}^\top \mathbf{s}_i)_l \right].$$

- Each competing event has its own Weibull pair (ν_l, ρ_l) .
- $\text{CIF}_{i,l}(t)$ gives the probability that cause l has occurred by t in the presence of competing causes.

Appendix: discrete event sampling in Leaspy

$$F_i^{\text{cond}}(t_{i,r}) = \begin{cases} 1 - S_i(t_{i,r})/S_i(t_{i,0}), & \text{single event,} \\ \sum_{l=1}^L \text{CIF}_{i,l}(t_{i,r})/S_i(t_{i,0}), & \text{competing events.} \end{cases}$$

- 1 Draw $U \sim \mathcal{U}[0, 1]$.
- 2 Select the first planned visit where $F_i^{\text{cond}}(t_{i,r}) \geq U$.
- 3 If no visit crosses U , right-censor at the last visit.
- 4 For competing risks, sample the cause from incremental cause-specific CIFs.

This ensures internal API reuse, but discretises event times onto the visit grid.

Appendix: detailed population recovery (1/2)

Parameter	θ^*	$\bar{\theta}$	RB (%) [95% CI]	RRMSE (%) [95% CI]	RSE [95% CI]
$g^{(0)}$	10.461	9.595	-8.27 [-9.41, -7.23]	11.41 [10.60, 12.28]	0.086 [0.076, 0.095]
$g^{(1)}$	2.326	2.122	-8.80 [-10.03, -7.52]	12.61 [11.55, 13.66]	0.099 [0.087, 0.111]
$g^{(2)}$	1.951	1.803	-7.60 [-8.71, -6.52]	11.16 [10.26, 12.06]	0.089 [0.079, 0.098]
$g^{(3)}$	3.415	3.213	-5.92 [-6.68, -5.17]	8.09 [7.46, 8.76]	0.059 [0.052, 0.066]
$v_0^{(0)}$	0.078	0.083	6.46 [5.32, 7.65]	10.53 [9.56, 11.53]	0.078 [0.070, 0.086]
$v_0^{(1)}$	0.257	0.263	2.23 [1.54, 2.93]	5.57 [5.05, 6.12]	0.050 [0.045, 0.054]
$v_0^{(2)}$	0.238	0.241	1.12 [0.52, 1.74]	4.52 [4.12, 4.92]	0.043 [0.039, 0.047]
$v_0^{(3)}$	0.133	0.135	2.05 [1.37, 2.73]	5.25 [4.75, 5.78]	0.048 [0.043, 0.052]

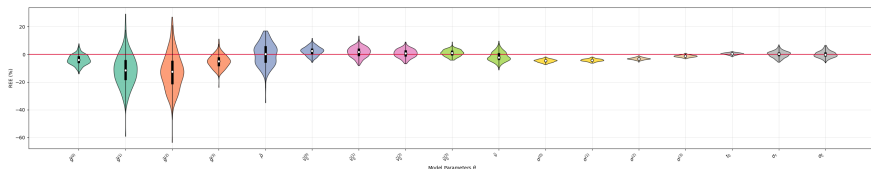
Intervals are 95% percentile bootstrap confidence intervals from 2,000 resamples of the $M = 200$ cohort-level estimates.

Appendix: detailed population recovery (2/2)

Parameter	θ^*	$\bar{\hat{\theta}}$	RB (%) [95% CI]	RRMSE (%) [95% CI]	RSE [95% CI]
ρ	1.849	1.851	0.10 [-0.61, 0.81]	5.01 [4.50, 5.51]	0.050 [0.045, 0.055]
ν	4.047	4.142	2.33 [1.64, 3.00]	5.43 [4.97, 5.91]	0.048 [0.043, 0.053]
$\sigma^{(0)}$	0.069	0.065	-4.60 [-4.76, -4.45]	4.73 [4.58, 4.89]	0.012 [0.011, 0.013]
$\sigma^{(1)}$	0.079	0.075	-4.16 [-4.28, -4.04]	4.25 [4.14, 4.38]	0.009 [0.009, 0.010]
$\sigma^{(2)}$	0.074	0.072	-3.08 [-3.17, -2.97]	3.17 [3.07, 3.27]	0.008 [0.007, 0.009]
$\sigma^{(3)}$	0.045	0.044	-1.26 [-1.37, -1.15]	1.47 [1.37, 1.56]	0.008 [0.007, 0.009]
t_0	56.699	56.948	0.44 [0.35, 0.53]	0.83 [0.76, 0.90]	0.007 [0.006, 0.008]
σ_τ	11.630	11.639	0.08 [-0.24, 0.38]	2.32 [2.10, 2.54]	0.023 [0.021, 0.025]
σ_ξ	1.078	1.079	0.03 [-0.27, 0.37]	2.32 [2.11, 2.56]	0.023 [0.021, 0.026]

θ^* is the generating value; $\bar{\hat{\theta}}$ is the mean estimate across the 200 independently fitted cohorts.

Appendix: population relative-error distributions



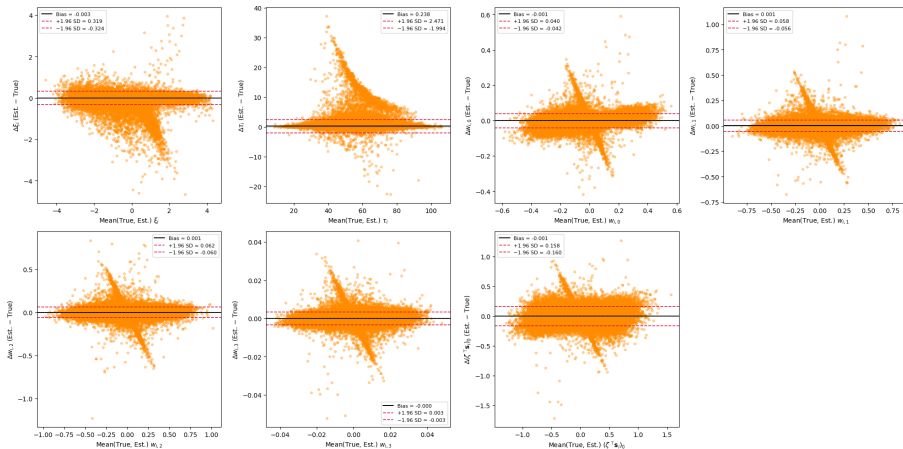
Red: zero relative error. Temporal parameters concentrate closest to zero; trajectory positions $g^{(j)}$ show the largest systematic underestimation.

Appendix: detailed individual-effect recovery

Effect	ICC(3,1)
ξ_i	0.988 ± 0.004
τ_i	0.995 ± 0.002
$w_{i,0}$	0.987 ± 0.006
$w_{i,1}$	0.989 ± 0.003
$w_{i,2}$	0.990 ± 0.003
$w_{i,3}$	0.987 ± 0.003
$(\zeta^\top \mathbf{s}_i)_0$	0.965 ± 0.027

Values are mean $\pm$ standard deviation across the $M = 200$ cohorts, as reported in the report.

Appendix: Bland–Altman diagnostic



Black: mean difference; red: limits of agreement. Biases are close to zero for all effects.

Simulation unit tests

- non-empty random-mode output
- expected patient count
- strictly increasing visit times
- DataFrame schedules are respected
- all model features are returned

estimate functional tests

- univariate numerical correctness
- raw/DataFrame consistency
- MultiIndex preservation
- multivariate output shape:
 $n_{\text{timepoints}} \times (J + L)$

Appendix: limitations and next experiments

What the reported experiment establishes

Self-consistency of the integrated pipeline for $N = 1000$ under a favourable, dense visit design.

What it does not establish

Universal recovery under sparse visits, heavier censoring, smaller cohorts, or model misspecification.

Perspectives stated in the report

Vary sample size, censoring intensity, visit sparsity, and misspecification; explore more flexible visit schedules and noise models; align continuous-time event generation more closely with the grid-based implementation.

Appendix: results of the original paper's simulation study

	Parameters name		RB (%)	RRMSE (%)
Distribution of random effects	Estimated reference time (mean)	t_0	0.10	2.23
	Estimated reference time (std)	σ_τ	-1.47	4.40
	Individual log-speed factor (std)	σ_ξ	13.21	14.85
Longitudinal fixed effects	Curve values at $t_0: \frac{1}{1+g_k}$	g_0	-5.35	9.28
		g_1	-8.08	11.27
		g_2	-7.74	10.87
		g_3	-5.93	7.71
	Speed of the logistic curves ($v_{0,k}$)	$v_{0,0}$	-7.04	10.87
		$v_{0,1}$	-7.43	10.64
		$v_{0,2}$	-8.89	11.27
		$v_{0,3}$	-9.34	11.21
	Estimated noises (σ_k)	σ_0	-3.22	4.12
		σ_1	-1.08	2.44
		σ_2	-0.82	2.24
		σ_3	0.17	2.14
Survival fixed effects	Weibull scale (ν_I)	ν_0	18.04	23.19
		ν_1	-0.09	9.97
	Weibull shape (ρ_I)	ρ_0	-8.38	14.15
		ρ_1	10.98	19.79

Results are extracted from: Ortholand et al. [5].